

Skills Nobody Taught: Unsupervised Skill Discovery for Manipulating Granular Media

Progress report — Transport-Metric Skill Discovery (TMSD)

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The story in one slide

1. **The dream:** robots that *invent* their own useful behaviors — no rewards, no demonstrations.
2. **The field:** three generations of methods (DIAYN $\rightarrow$ DADS $\rightarrow$ METRA), each fixing the last one's flaw.
3. **The gap:** none of them can manipulate *stuff* — soil, dough, sand. They learn to move *themselves*.
4. **Our idea:** measure skill diversity by the *physical cost of moving material* (optimal transport / Wasserstein), computed *only* on the material.
5. **What we found:** half our hypothesis died honestly, the other half is the strongest result — and the post-mortem produced a unifying theory of *why* each method family fails on materials.
6. **The demo:** an excavator that taught itself to dig, and executes “dig here, dump there” orders it was never trained on.

Roadmap

Background: teaching robots without teaching

Three generations of methods, three definitions of “different”

Our idea: measure diversity in units of moved material

The experimental campaign, step by step

The full comparison

Demonstration

Where this goes

Background: teaching robots without teaching

What is unsupervised skill discovery?

Ordinary RL: an engineer writes a reward function (+1 for a full bucket, -0.01 per second. . .), the robot maximizes it. Every new task $\Rightarrow$ new reward $\Rightarrow$ new training.

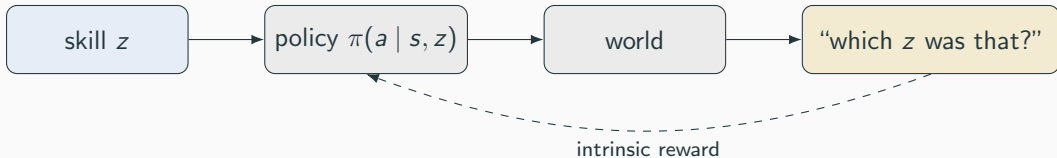
Unsupervised skill discovery (USD): give the robot *no reward at all*. Instead give it a “skill knob” z (a vector it is handed at the start of each attempt) and one meta-instruction:

“Behave so differently for different z that an observer could tell which z you were given.”

Like a child alone in a sandbox: nobody defines “dig” or “pile” — distinct, repeatable behaviors emerge from the drive to do *distinguishable things*. Afterwards, the skill knob becomes a *control interface*: point z at what you want.

The formal skeleton (shared by every method)

- Skill $z \sim p(z)$ fixed for an episode; policy $\pi(a \mid s, z)$.
- Learn π to maximize an **intrinsic** (self-generated) reward measuring how *informative about z* the visited states are.
- The methods differ in exactly one thing: **what “different” means**.



**Three generations of methods,
three definitions of “different”**

Generation 1 — DIAYN (2019): “be identifiable”

Idea. Maximize the mutual information $I(S; Z)$: train a discriminator $q(z \mid s)$ that guesses the skill from the state; reward the policy when the guess is easy.

$$r = \log q(z \mid s) - \log p(z)$$

What emerges. Distinct poses and gaits — genuinely different behaviors, zero supervision. Landmark result.

The flaw. *Identifiable $\neq$ substantial.* A skill is perfectly identifiable if it tilts one joint by 1° and holds still. MI is maximized by **microscopic, static differences** — there is no pressure to *keep changing* the world. (We will meet this flaw again, measured, on soil.)

Generation 2 — DADS (2020): “be predictable”

Idea. Skills should have consequences you can *model*. Learn skill-conditioned dynamics $q(s' | s, z)$ and reward transitions that are predictable under your own z but unlike other skills' transitions:

$$r = \log q(s' | s, z) - \log \frac{1}{L} \sum_l q(s' | s, z_l)$$

What emerges. Smooth, model-friendly locomotion primitives; enables planning in skill space.

The flaw. The signal lives in **per-step state changes**. If the interesting part of the world changes rarely or slowly, every skill's next state is equally (perfectly) predictable — the two terms cancel and the reward **evaporates**. (We measured exactly this on soil: reward $\equiv 0.000$ for 200,000 steps.)

Generation 3 — METRA (2024): “go far”

Idea. Stop asking “identifiable?” and ask “*far apart?*”. Learn a map ϕ of states into a latent space and make skills travel in different latent directions:

$$\max_{\pi, \phi} \mathbb{E}[(\phi(s') - \phi(s))^{\top} z] \quad \text{s.t.} \quad \|\phi(s) - \phi(s')\| \leq d(s, s'),$$

where d is a distance between states (METRA uses *temporal* distance: how many steps to get from one state to the other). The constraint stops ϕ from cheating by inflating distances.

Key property. The per-step rewards *telescope*:

$$\sum_t (\phi(s_{t+1}) - \phi(s_t))^{\top} z = (\phi(s_T) - \phi(s_0))^{\top} z.$$

Total reward = **net displacement**. You cannot earn it by micro-wiggling; you must *cumulatively change the world*. State of the art on locomotion.

The remaining flaw...

The gap all three share: they move themselves, not stuff

The state s contains everything — joint angles, velocities, *and* the world outside. The richest, easiest-to-vary part of s is the robot's **own body**. All three objectives happily feast on it.

“Previous approaches only learn to gain control of the agent’s own body, completely ignoring the object.” — the CSD authors (ICML 2023), on DIAYN/LSD in a manipulation task

And nobody has tried materials. Skill discovery is confined to rigid bodies and locomotion. Meanwhile the hard, valuable frontier of robotics is *deformable and granular matter*: soil, dough, sand, grain — where writing rewards by hand is hardest, i.e. exactly where USD would matter most.

Open question we attacked: *can a robot invent manipulation of a material — digging, piling, shaping — with zero supervision?*

**Our idea: measure diversity in units
of moved material**

Wasserstein distance, intuitively: the earth mover

Suppose the world state is a *pile of sand*, described by how much mass sits at each location. How different are two piles?



The **Wasserstein distance** $W_2(\mu, \nu)$ is the minimum total *work* — mass $\times$ distance — a perfect bulldozer would need to convert one pile into the other. It is also called the *earth mover's distance*, and for once the metaphor is literal: **our robot is the earth mover**.

Contrast: a pixel-by-pixel (Euclidean) comparison calls two piles “very different” even when one is just the other shifted 10 cm — it has no idea that moving mass *far* costs more than moving it *near*.

Wasserstein distance, mathematically

For probability measures μ, ν on $\mathbb{R}^n$ (Kantorovich formulation):

$$W_2(\mu, \nu) = \left(\min_{\gamma \in \Pi(\mu, \nu)} \int \|x - y\|^2 d\gamma(x, y) \right)^{1/2},$$

where $\Pi(\mu, \nu)$ is the set of *transport plans* (joint distributions with marginals μ, ν): $\gamma(x, y)$ says how much mass travels from x to y .

The 1D gift. On a line, the optimal plan never lets mass streams cross, so it is given by matching quantiles. With CDFs F_μ, F_ν and quantile functions F^{-1} :

$$W_2^2(\mu, \nu) = \int_0^1 (F_\mu^{-1}(u) - F_\nu^{-1}(u))^2 du.$$

Closed form. No optimization. Microseconds on a GPU. Our soil bed is (to first order) a 1D mass profile over x — so we get the exact physical metric essentially for free, thousands of times per second during training.

Transport-Metric Skill Discovery (TMSD): the two ingredients

Take METRA's template and change *two things*:

1. **Physical ground metric.** Set $d(s, s') = W_2(\mu_s, \mu_{s'})$ — the transport cost between the *soil mass distributions*. Latent displacement is then bounded by *tons-moved-times-meters*:

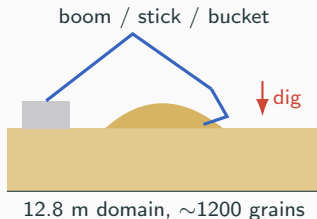
$$\max_{\pi, \phi} \mathbb{E}[(\phi(\mu') - \phi(\mu))^{\top} z] \quad \text{s.t.} \quad \|\phi(\mu) - \phi(\mu')\| \leq W_2(\mu, \mu').$$

2. **External-state conditioning.** ϕ sees *only the soil* — never a joint angle, never a velocity. A skill that merely waves the arm produces $\phi(\mu') = \phi(\mu)$: **reward exactly zero, by construction**.

Hypothesis going in: ingredient 1 (the metric) is the headline, ingredient 2 is plumbing.
Reality (spoiler): precisely backwards — and proving that took 33 training runs.

Testbed: a 2D excavator on granular physics

- Custom DEM simulation: ~ 1200 soil grains, Hertz contacts, friction, cohesion; CUDA backend.
- 3-joint hydraulic arm (boom/stick/bucket) with realistic torque limits; fixed base.
- Observation: proprioception + soil heightfield. Action: 3 joint velocities.
- Soil measure μ : mass histogram over x (64 bins), **conserved** $\Rightarrow$ balanced optimal transport is well-posed.
- Skills: continuous z on the unit sphere, $\dim = 4$.

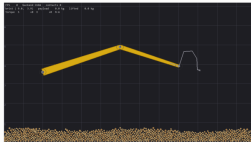


Everything below runs on **one desktop GPU** (RTX 5080): 33 runs, ≈ 70 GPU-hours

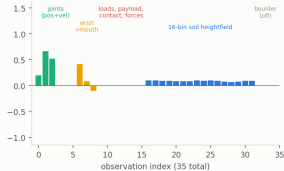
Who sees what: the four-layer state representation

One state, four representations: from physics to the transport map

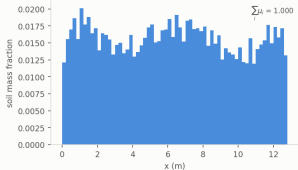
(a) The world: ~1200 simulated grains + 3-joint arm



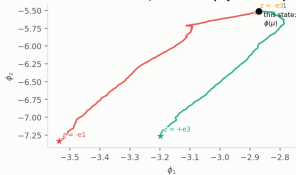
(b) What the POLICY sees: body + coarse soil (35-dim)



(c) What the OBJECTIVE sees: $\mu = 64$ -bin soil mass distribution -- zero robot information, sums to 1



(d) The learned map: $\phi(\mu) \in \mathbb{R}^d$ (first 2 dims)
skills travel directions; distance $\approx$ physical transport



The policy (b) may know its body — it must, to control it. The *objective's* representation ϕ sees only (c): the soil mass distribution, zero robot information. (d): the learned latent map, with real skill episodes traced. The paper's central finding lives entirely in the split between (b) and (c)

The experimental campaign, step by step

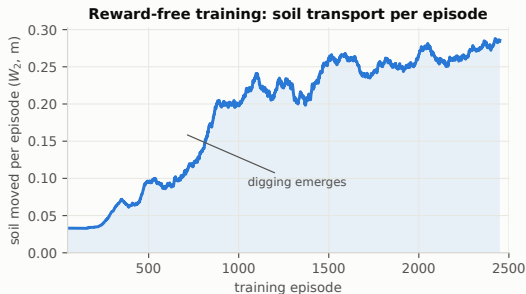
Experiment 1 — Does anything emerge at all?

Problem. Nobody has shown reward-free skills on granular media. Does the objective even produce soil interaction?

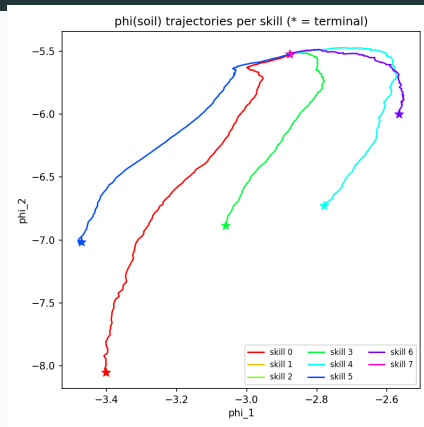
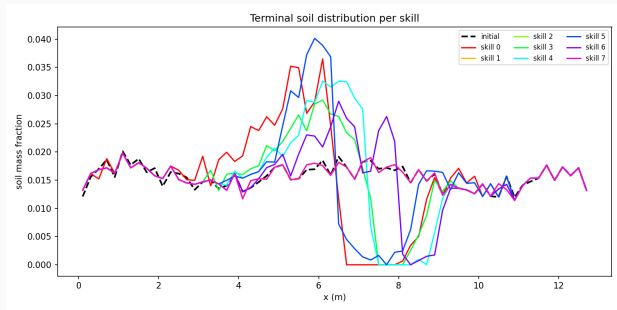
Test. 500k steps, fixed terrain, TMSD objective (W_2 metric, soil-only ϕ).

Result. **Yes.** Digging, piling, and directed mass transport emerge; late episodes move ~ 0.27 m of soil-mass (in W_2), $8\times$ the starting level, with **zero** rewards.

Insight. Skill coordinate maps to physics: the sign of z_1 literally equals the direction the soil's center of mass shifts. The skill space *is* a transport control knob.



Experiment 1 — what the skills actually do to the soil



Left: terminal soil profile per skill vs. the initial terrain (dashed) — different skills dig and pile at different places. **Right:** the learned latent map $\phi(\text{soil})$: skill trajectories fan out in distinct directions — the “mental map” of soil states, where distance $\approx$ tons moved $\times$ meters.

Experiment 2 — Is the Wasserstein metric doing the work?

Problem. Our headline claim: the *physical* metric matters. A reviewer's first attack: "this is just METRA with a different d ."

Test. Same recipe (random terrain, 3 seeds each), swap only the ground metric: W_2 vs. Euclidean-on-histograms vs. METRA's temporal $d \equiv 1$. Judge all by the same physical yardstick.

ground metric	W_2 (ours)	Euclidean	temporal (METRA)
skill coverage (m, mean of 3 seeds)	0.326 ± 0.036	0.352 ± 0.034	0.291 ± 0.094

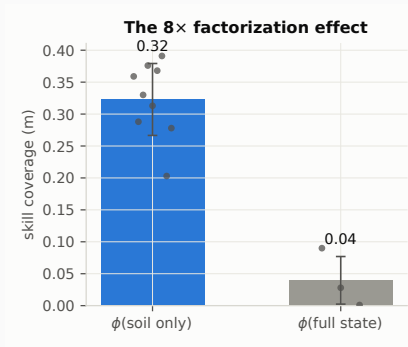
Result. A **three-way tie**. The metric hypothesis is **falsified** — our own pre-registered ablation killed our own headline in one day.

Insight (the good part). Every trained ϕ — regardless of metric — ended up 0.99-correlated with physical W_2 . The reachable soil manifold is low-dimensional; on it, *all sensible metrics agree*. Geometry is **emergent, free, and metric-agnostic**.

Experiment 3 — Then what IS doing the work?

Problem. If not the metric — is it the *conditioning* (soil-only ϕ)?

Test. Identical run, but let ϕ see the full state (proprioception + soil). This *is* vanilla METRA, faithfully. 3 seeds.



Result. 8 $\times$ effect. Full-state ϕ : 7/8 skills never touch the soil; the arm performs

Experiment 4 — Give the metric a fair second chance

Problem. Maybe the 1D soil profile was too crude for metrics to differ. On a full 2D mass grid (deep-narrow vs. wide-shallow holes now distinct), transport structure should finally matter.

Test. 2D measure (48×24 mass grid), sliced- W_2 (closed-form random 1D projections) vs. Euclidean vs. temporal.

Result. *Worse than falsified.* Sliced- W_2 : coverage 0.044 — **$5\times$ worse than Euclidean (0.245) on its own ruler.**

Insight (mechanism, from telemetry). Not misbehavior — *starvation*: under the SW_2 constraint the representation extracted only $\sim 7\%$ of its allowed reward budget (Euclidean control: 45%), so the policy never received a usable gradient. The physical transport metric did not mis-price soil work; it **failed to feed learning**. Honest open question, one paragraph in the paper.

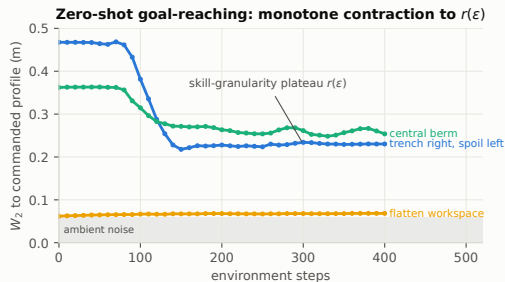
Experiment 5 — Can the skills reach goals? (the payoff layer)

Problem. Diverse skills are only useful if they compose into “make the soil look like *this*.”

Test. Zero-shot rule, no goal training: repeatedly command $z = \frac{\phi(\mu^*) - \phi(\mu_t)}{\|\cdot\|}$ toward target profile μ^* . 21 targets, all trained models.

Result. 39–63% of the distance-to-goal removed, zero-shot; 79–94% of control intervals *monotonically* decrease W_2 — an empirical **contraction certificate**.

Insight. Curves plateau at ≈ 0.23 m: the *skill-granularity radius* $r(\epsilon)$ our certificate theory predicts. Skills contract to a ball around the goal, not to a point — this number will explain a later failure



Experiments 6–7 — The other families, same arena (DIAYN, DADS)

Problem. Is *conditioning* alone the whole story? Then DIAYN- and DADS-style objectives, given soil-only inputs, should also work.

Test. Full 3×2 grid: objective family $\times$ conditioning, 2–3 seeds per cell, identical recipe and frozen evaluation.

objective family	soil-only	full-state (vanilla)
distance-max (METRA-style, any metric)	0.29–0.35 ✓	0.039 ✗
MI discriminator (DIAYN-style)	0.021 ✗	0.031 ✗
skill dynamics (DADS-style)	0.008 ✗	0.130* ✗

Result. Only one cell works. Neither ingredient suffices alone; both are necessary.

Mechanisms (measured, not guessed):

- **DIAYN+soil saturates:** discriminator hits 0.98 cosine accuracy off *microscopic* grain differences — identifiable without moving anything (Gen-1 flaw on soil)

The unifying insight: materials are slow states

Soil changes rarely and slowly; the arm changes richly at every step. This one property explains the whole grid:

MI (DIAYN)

per-step *identifiability*

⇒ **saturates** on
microscopic differences

dynamics (DADS)

per-step *predictability*

⇒ **starves**: slow states
trivially predictable

distance-max (METRA)

reward *telescopes* to net
terminal displacement

⇒ **must accumulate change**

$$\underbrace{\sum_t (\phi(s_{t+1}) - \phi(s_t))^T z}_{\text{what the policy collects}} = \underbrace{(\phi(s_T) - \phi(s_0))^T z}_{\text{net change of the material}}$$

Design law: *on slow external states, the objective must (i) integrate change over time, not score it per step, and (ii) see only that state while doing so.* Each frontier family lacks one of the two.

Experiment 8 — Stress test: arbitrary user goals (where it breaks)

Problem. Interactive demo: user sculpts terrain freehand, records it, wrecks it, robot restores. Reality: *it failed, badly.*

Diagnosis chain (each step measured):

1. Off-distribution states? Retrained *on* user-like edits → *no change.*
 2. Bad skill *selection*? Built a planner with **perfect lookahead** (model-predictive control using the simulator itself as the model) → *+6–13% only.*
- ⇒ With perfect selection ruled out, the **vocabulary itself** is the limit: $\sim 3\text{--}5$ coarse transport modes cannot express local repairs. And user goals sit *inside* the $r(\varepsilon) \approx 0.2\text{ m}$ granularity radius from Exp. 5 — the method performs exactly to its certificate, and the certificate says “too coarse for this.”

Insight (the honest headline): **diversity $\neq$ controllability.** Skill discovery finds the few *most mutually different* behaviors, not a dense basis of all behaviors. Its proper role — consistent with how the literature quietly uses it — is *pretraining and coarse*

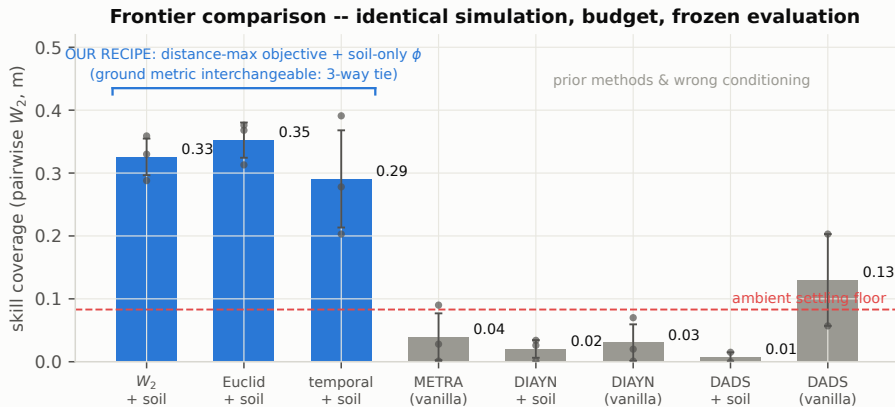
The full comparison

Every method, one arena, one frozen ruler

33 runs, ≈ 70 GPU-hours, identical simulation/recipe/budget (200k steps unless noted), evaluation protocol frozen *before* results existed. Coverage = mean pairwise W_2 between terminal soil states (8 fixed skills $\times$ 3 fixed terrains).

method	lineage	coverage (m)	soil/ep (m)	seeds
distance-max + soil-only ϕ , W_2 (TMSD)	ours	0.326 ± 0.036	0.21	3
distance-max + soil-only ϕ , Euclidean	ablation	0.352 ± 0.034	0.21	3
distance-max + soil-only ϕ , temporal	METRA+our fix	0.291 ± 0.094	0.21	3
distance-max, full-state (= vanilla METRA)	frontier	0.040 ± 0.046	0.05	3
MI, soil-only (DIAYN + our fix)	frontier+fix	0.021 ± 0.015	0.04	3
MI, full-state ($\approx$ vanilla DIAYN)	frontier	0.031 ± 0.028	0.05	3
dynamics, soil-only (DADS + our fix)	frontier+fix	0.008 ± 0.008	0.04	2
dynamics, full-state ($\approx$ vanilla DADS)	frontier	0.130 ± 0.073	0.11	2
sliced- W_2 on 2D measure	ours (negative)	0.044	0.04	1
goal-conditioned SAC from scratch	control exp.	— (fails to learn)	—	1

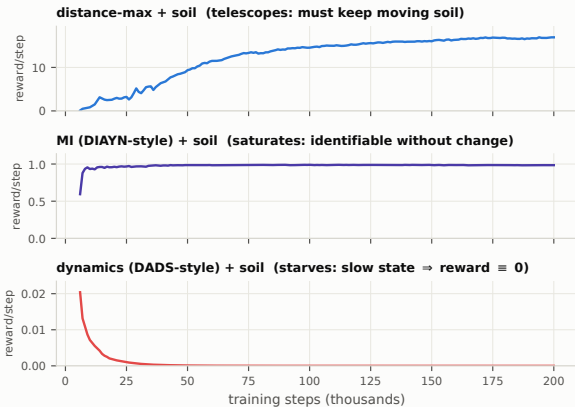
The same table as a picture



Three blue towers, five gray stumps: the towers share both ingredients; every stump is missing at least one. Dots are individual seeds.

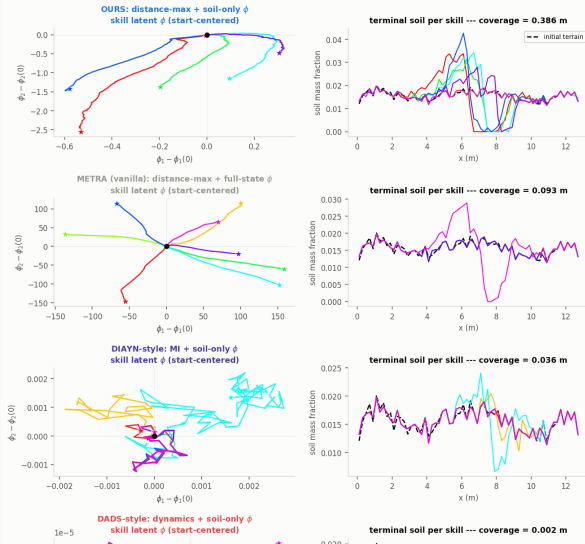
Mechanisms, watched during training (real curves)

Three objective families on the same slow material state



Inside the latent spaces: the taxonomy as four axis scales

Same 8 skills, same terrain: each method's latent map vs. what its skills physically do to the soil



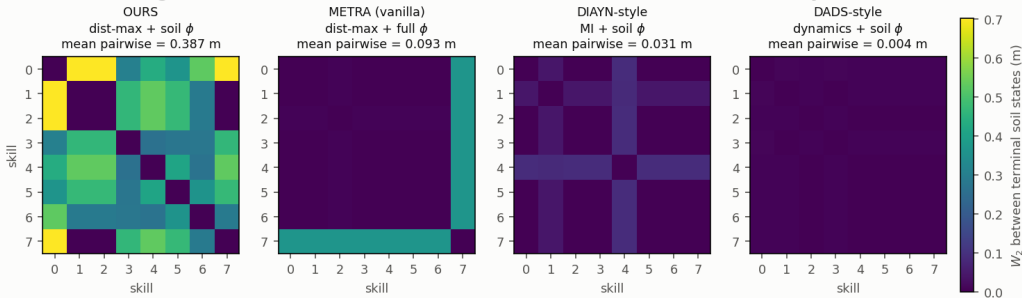
Same 8 skills, same terrain, each method's own latent map (left) vs. what its skills do to the soil (right).

	latent spread	coverage
METRA	~ 150	0.09 m
ours	~ 2.5	0.39 m
DIAYN	~ 0.002	0.04 m
DADS	~ 0.00007	0.00 m

METRA: a beautiful ± 150 latent fan above *untouched soil* — diversity decoupled from the world. Ours: the only row where latent geometry and soil geometry agree, because the W_2 cap

Coverage matrices: the fingerprint view (one shared color scale)

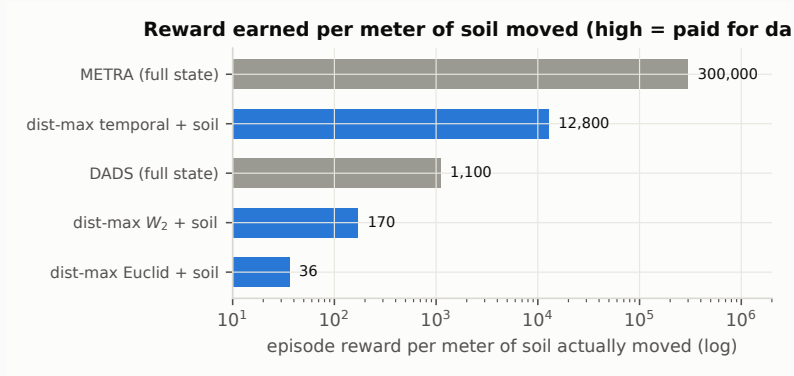
Coverage matrices, one shared color scale: same 8 skills, same terrain, every method



Bright cell = that skill pair left the soil in genuinely different states. Ours: a quilt. METRA: one bright cross — a single working skill, seven latent-diverse clones. DIAYN: a ghost. DADS: black. The frontier table's coverage numbers are the means of these matrices.

What each objective is actually paid for (mechanism fingerprints)

Episode reward earned per meter of soil actually moved (log scale) — a lie detector for objectives:



The W_2 constraint's one surviving virtue: reward in **physical units** — the value function is denominated in meters of mass transport. Full state models earn $\sim 2000\times$ more reward per

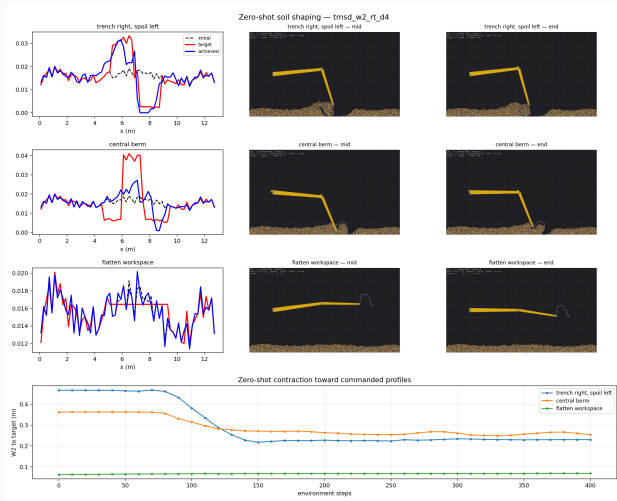
Demonstration

The demonstration (live, 3 acts, ~2.5 min, one script)

1. **“Nobody taught it to dig.”** Four discovered skills cycle on random terrain: dig-and-pile-right, carve-left, excavate-middle, wide-cut. Zero rewards, zero demonstrations, captioned live.
2. **“Without our ingredient.”** Split screen, same terrain, same training: soil-only ϕ digs; full-state ϕ (= vanilla METRA) performs elaborate arm choreography over **untouched soil**. The $8\times$ result, visible in one frame.
3. **“Give it a job.”** A red target line appears: *dig the right zone, dump left*. Zero-shot skill steering executes:
79% of the ordered zone’s soil removed; dump zone filled to within 2.6 points of target; W_2 to goal halved — live counter, before/after cards.

Also available interactively: sculpt terrain with the mouse, record a goal state, wreck it, press ENTER — with the honest caveat that free-form restoration sits at the method’s granularity limit (Exp. 8), which we state on-screen rather than hide.

Zero-shot shaping, in full (sim frames + profiles + contraction)



Where this goes

Contributions (what we can claim today)

1. **First reward-free skill discovery on granular media** — digging/piling/shaping emerge with zero supervision (gap confirmed against literature).
2. **The two-ingredient law**, with an $8\times$ /complete-failure contrast across a 3×2 grid, 2–3 seeds per cell: *cumulative* objectives + *external-only* conditioning; each necessary, neither sufficient.
3. **Metric-agnosticism** (+ its explanation): on low-dimensional material manifolds, all Lipschitz ground metrics yield the same skills; latent geometry is emergent (0.99 isometry to physical W_2 , unasked). A measured counterpoint to the METRA-lineage emphasis — and to our own initial bet.
4. **Certified zero-shot shaping**: 39–63% goal-distance reduction, $\sim 90\%$ monotone contraction, plateau = predicted $r(\varepsilon)$.
5. **Slow-state failure taxonomy**: saturation (MI) / starvation (dynamics) / telescoping (distance-max) — a design law for material-manipulation objectives.

Limits (stated plainly) and next steps

Limits.

- Skill granularity $r(\varepsilon) \approx 0.2$ m: coarse shaping yes, precision restoration no. Diversity $\neq$ controllability.
- Fixed machine base $\Rightarrow$ soil transports only *toward* the excavator (like a real backhoe that cannot drive).
- One domain family, one simulator, 1D-dominant measure; sliced- W_2 starvation unexplained.

Next.

- **Second material (dough/plasticine)** — the generality gate: same objective, unchanged.
- **Mobile base + warm-started goal-conditioned layer** — closes the reachability and precision gaps for deployment-grade demos.
- **Contraction theorem** — formalize: ε -coverage of transport directions $\Rightarrow$

**An excavator taught itself to dig.
We measured why that worked —
and why nothing else did.**

Demo: `python scripts/showcase.py`